In [ ]:
```
Experiment 8: Data Pre-processing & Feature Engineering
PIN: 241UDS0101

Aim:-
To implement data pre-processing and feature engineering on a student
dataset by removing unnecessary features, scaling numerical features, and
encoding categorical variables using Python libraries.
```

In [ ]:
```
## Step 1: Import the basic libraries
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
## Step 2: Load & Analyze the Dataset
```

In [2]:
```
df_241UDS0101 = pd.read_csv("Experiment_8_241UDS0101_Student_Features.csv")

# Dataset information
df_241UDS0101.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 121 entries, 0 to 120
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Branch                  120 non-null    object 
 1   Marks[10th]             120 non-null    float64
 2   Marks[12th]             120 non-null    float64
 3   Gender                  117 non-null    object 
 4   Board[10th]             117 non-null    object 
 5   Board[12th]             117 non-null    object 
 6   Category                117 non-null    object 
 7   GPA 1                   117 non-null    float64
 8   Rank                    120 non-null    float64
 9   Normalized Rank         120 non-null    float64
 10  CGPA                    117 non-null    float64
 11  Current Back            120 non-null    float64
 12  Ever Back               117 non-null    float64
 13  GPA 2                   120 non-null    float64
 14  GPA 3                   120 non-null    float64
 15  GPA 4                   120 non-null    float64
 16  GPA 5                   120 non-null    float64
 17  GPA 6                   120 non-null    float64
 18  Olympiads Qualified     120 non-null    float64
 19  Technical Projects      117 non-null    float64
 20  Tech Quiz               120 non-null    float64
 21  Engg. Coaching          117 non-null    float64
 22  NTSE Scholarships       120 non-null    float64
 23  Miscellany Tech Events  117 non-null    float64
dtypes: float64(19), object(5)
memory usage: 22.8+ KB
```

In [3]:
```
df_241UDS0101.head()
```

Out[3]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	CIVIL	93.14	58.56	NaN	RBSE	CBSE	GEN
1	ECE	99.00	67.69	Female	State Board	RBSE	OBC
2	IT	92.04	70.88	Female	ICSE	BSEB Patna	OBC
3	CIVIL	83.68	68.49	Female	State Board	BSEB Patna	OBC
4	ECE	74.54	86.17	Male	CBSE	CBSE	ST

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	7.60	13991.0	24.603702	...	5.60	5.86	6.90
1	6.50	52283.0	77.010434	...	6.17	7.40	7.04
2	6.79	36061.0	93.795547	...	7.06	8.16	6.64
3	5.44	39497.0	41.101897	...	8.38	7.12	8.10
4	5.86	41061.0	29.481151	...	6.33	5.58	7.14

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	7.34	0.0	1.0	4.0	0.0	0.0	0.0
1	6.61	0.0	3.0	2.0	0.0	0.0	0.0
2	8.19	0.0	1.0	2.0	1.0	0.0	0.0
3	7.33	0.0	1.0	4.0	0.0	0.0	0.0
4	8.34	0.0	3.0	5.0	1.0	0.0	0.0

5 rows × 24 columns

In [4]:
```
df_241UDS0101.shape
```

Out[4]:
```
(121, 24)
```

In [5]:
```
# Check for null values
df_241UDS0101.isnull().sum()
```

Out[5]:
```
Branch                   1
Marks[10th]              1
Marks[12th]              1
Gender                   4
Board[10th]              4
Board[12th]              4
Category                 4
GPA 1                    4
Rank                     1
Normalized Rank          1
CGPA                     4
Current Back             1
Ever Back                4
GPA 2                    1
GPA 3                    1
GPA 4                    1
GPA 5                    1
GPA 6                    1
Olympiads Qualified      1
Technical Projects       4
Tech Quiz                1
Engg. Coaching           4
NTSE Scholarships        1
Miscellany Tech Events   4
dtype: int64
```

In [6]:
```
# Remove rows that are completely empty
df_241UDS0101 = df_241UDS0101.dropna(how="all").copy()
df_241UDS0101.shape
```

Out[6]:
```
(120, 24)
```

In [ ]:
```
## Step 3: Remove Unnecessary Features
The `Rank` column is removed, following the experiment procedure.
```

In [7]:
```
df_241UDS0101 = df_241UDS0101.drop("Rank", axis=1)
df_241UDS0101.shape
```

Out[7]:
```
(120, 23)
```

In [ ]:
```
## Step 4: Scale the Numerical Features
Use `StandardScaler` on `Marks[10th]` and `Marks[12th]`. Then use
`MinMaxScaler` on `CGPA` and `GPA 1`. Missing numeric values in these
columns are filled with their median first so scaling can run successfully.
```

In [8]:
```
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, MinMaxScaler

# Fill missing values in the columns to be scaled
scaled_columns_241UDS0101 = [
    "Marks[10th]", "Marks[12th]", "CGPA", "GPA 1"
]
imputer_scaled_241UDS0101 = SimpleImputer(strategy="median")
df_241UDS0101[scaled_columns_241UDS0101] =
imputer_scaled_241UDS0101.fit_transform(
    df_241UDS0101[scaled_columns_241UDS0101]
)

# Standardize 10th and 12th marks
z_scaler_241UDS0101 = StandardScaler()
df_241UDS0101[["Marks[10th]", "Marks[12th]"]] =
z_scaler_241UDS0101.fit_transform(
    df_241UDS0101[["Marks[10th]", "Marks[12th]"]]
)
df_241UDS0101.head(3)
```

Out[8]:

Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	CIVIL	1.604414	-1.336212	NaN	RBSE	CBSE	GEN
1	ECE	2.267175	-0.545689	Female	State Board	RBSE	OBC
2	IT	1.480005	-0.269482	Female	ICSE	BSEB Patna	OBC

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
0	7.60	24.603702	6.89	...	5.60	5.86	6.90
1	6.50	77.010434	7.74	...	6.17	7.40	7.04
2	6.79	93.795547	6.78	...	7.06	8.16	6.64

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	7.34	0.0	1.0	4.0	0.0	0.0	0.0
1	6.61	0.0	3.0	2.0	0.0	0.0	0.0
2	8.19	0.0	1.0	2.0	1.0	0.0	0.0

3 rows × 23 columns

In [9]:
```
# Apply Min-Max normalization to CGPA and GPA 1
mm_scaler_241UDS0101 = MinMaxScaler()
df_241UDS0101[["CGPA", "GPA 1"]] = mm_scaler_241UDS0101.fit_transform(
    df_241UDS0101[["CGPA", "GPA 1"]]
)
df_241UDS0101.head(3)
```

Out[9]:

Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	CIVIL	1.604414	-1.336212	NaN	RBSE	CBSE	GEN
1	ECE	2.267175	-0.545689	Female	State Board	RBSE	OBC
2	IT	1.480005	-0.269482	Female	ICSE	BSEB Patna	OBC

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
0	0.602326	24.603702	0.513761	...	5.60	5.86	6.90
1	0.346512	77.010434	0.708716	...	6.17	7.40	7.04
2	0.413953	93.795547	0.488532	...	7.06	8.16	6.64

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	7.34	0.0	1.0	4.0	0.0	0.0	0.0
1	6.61	0.0	3.0	2.0	0.0	0.0	0.0
2	8.19	0.0	1.0	2.0	1.0	0.0	0.0

3 rows × 23 columns

In [ ]:
```
## Step 5: Encode the Categorical Features
Use `LabelEncoder` for `Gender`, `Category`, and `Branch`. Use
`pd.get_dummies()` for `Board[10th]` and `Board[12th]`, with
`drop_first=True`. Finally, convert Boolean dummy columns to integers.
```

In [10]:
```
from sklearn.preprocessing import LabelEncoder

# Fill missing categorical values before encoding
categorical_fill_columns_241UDS0101 = [
    "Gender", "Category", "Branch", "Board[10th]", "Board[12th]"
]
for col in categorical_fill_columns_241UDS0101:
    df_241UDS0101[col] = df_241UDS0101[col].fillna(df_241UDS0101[col].mode()
[0])

encoder_241UDS0101 = LabelEncoder()
for col in ["Gender", "Category", "Branch"]:
    df_241UDS0101[col] =
encoder_241UDS0101.fit_transform(df_241UDS0101[col])

df_241UDS0101.head(3)
```

Out[10]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	0	1.604414	-1.336212	1	RBSE	CBSE	1
1	2	2.267175	-0.545689	0	State Board	RBSE	2
2	4	1.480005	-0.269482	0	ICSE	BSEB Patna	2

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
0	0.602326	24.603702	0.513761	...	5.60	5.86	6.90
1	0.346512	77.010434	0.708716	...	6.17	7.40	7.04
2	0.413953	93.795547	0.488532	...	7.06	8.16	6.64

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	7.34	0.0	1.0	4.0	0.0	0.0	0.0
1	6.61	0.0	3.0	2.0	0.0	0.0	0.0
2	8.19	0.0	1.0	2.0	1.0	0.0	0.0

3 rows × 23 columns

In [11]:
```
# One-hot encode the board columns
df_241UDS0101 = pd.get_dummies(
    df_241UDS0101,
    columns=["Board[10th]", "Board[12th]"],
    drop_first=True
)
df_241UDS0101.head(3)
```

Out[11]:

Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Category	GPA 1	Normalized Rank
0	0	1.604414	-1.336212	1	1	0.602326	24.603702
1	2	2.267175	-0.545689	0	2	0.346512	77.010434
2	4	1.480005	-0.269482	0	2	0.413953	93.795547

Columns 9–15 of 22 · first column repeated

	CGPA	Current Back	Ever Back	...	NTSE Scholarships	Miscellany Tech Events	Board[10th]_CBSE
0	0.513761	0.0	0.0	...	0.0	0.0	False
1	0.708716	0.0	0.0	...	0.0	0.0	False
2	0.488532	0.0	0.0	...	0.0	0.0	False

Columns 16–22 of 22 · first column repeated

	Board[10th]_ICSE	Board[10th]_RBSE	Board[10th]_State Board	Board[12th]_CBSE	Board[12th]_ICSE	Board[12th]_RBSE	Board[12th]_State Board
0	False	True	False	True	False	False	False
1	False	False	True	False	False	True	False
2	True	False	False	False	False	False	False

3 rows × 29 columns

In [12]:
```
# Convert Boolean columns to integer values
bool_columns_241UDS0101 =
df_241UDS0101.select_dtypes(include="bool").columns
for col in bool_columns_241UDS0101:
    df_241UDS0101[col] = df_241UDS0101[col].astype(int)

df_241UDS0101.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 120 entries, 0 to 119
Data columns (total 29 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Branch                   120 non-null    int64
 1   Marks[10th]              120 non-null    float64
 2   Marks[12th]              120 non-null    float64
 3   Gender                   120 non-null    int64
 4   Category                 120 non-null    int64
 5   GPA 1                    120 non-null    float64
 6   Normalized Rank          120 non-null    float64
 7   CGPA                     120 non-null    float64
 8   Current Back             120 non-null    float64
 9   Ever Back                117 non-null    float64
 10  GPA 2                    120 non-null    float64
 11  GPA 3                    120 non-null    float64
 12  GPA 4                    120 non-null    float64
 13  GPA 5                    120 non-null    float64
 14  GPA 6                    120 non-null    float64
 15  Olympiads Qualified      120 non-null    float64
 16  Technical Projects       117 non-null    float64
 17  Tech Quiz                120 non-null    float64
 18  Engg. Coaching           117 non-null    float64
 19  NTSE Scholarships        120 non-null    float64
 20  Miscellany Tech Events   117 non-null    float64
 21  Board[10th]_CBSE         120 non-null    int64
 22  Board[10th]_ICSE         120 non-null    int64
 23  Board[10th]_RBSE         120 non-null    int64
 24  Board[10th]_State Board  120 non-null    int64
 25  Board[12th]_CBSE         120 non-null    int64
 26  Board[12th]_ICSE         120 non-null    int64
 27  Board[12th]_RBSE         120 non-null    int64
 28  Board[12th]_State Board  120 non-null    int64
dtypes: float64(18), int64(11)
memory usage: 28.1 KB
```

In [13]:
```
# Final dataset shape and missing-value check
print("Final shape:", df_241UDS0101.shape)
print("Total missing values:", int(df_241UDS0101.isnull().sum().sum()))
```

```
Final shape: (120, 29)
Total missing values: 12
```

In [14]:
```
# Save the processed dataset
df_241UDS0101.to_csv("processed_student_dataset_241UDS0101.csv",
index=False)
print("Saved: processed_student_dataset_241UDS0101.csv")
```

```
Saved: processed_student_dataset_241UDS0101.csv
```

In [ ]:
```
## Result
The student dataset is pre-processed by removing the unnecessary `Rank`
feature, scaling numerical features using `StandardScaler` and
`MinMaxScaler`, and encoding categorical variables using Label Encoding and
One-Hot Encoding. Boolean columns are converted to integers and the
processed dataset is saved as a CSV file.
```